

Statistics for Data Science: A Practical Learning Guide

Coursera-style Demo Course Material (Original, Copyright-Free)

Course Overview

This guide is original study material created for AI Tutor/RAG demonstrations. It is not affiliated with Coursera.

Learning Objectives

Learn descriptive statistics, probability, normal distribution, Central Limit Theorem, confidence intervals and hypothesis testing.

Module 1: Introduction to Statistics

Statistics is the science of collecting, organizing, analyzing and interpreting data.

Module 2: Population vs Sample

Population is the complete dataset. A sample is a subset selected for analysis.

Module 3: Measures of Central Tendency

Mean, Median and Mode summarize the center of data.

Module 4: Variance & Standard Deviation

Variance measures spread; standard deviation is its square root.

Module 5: Normal Distribution

The normal distribution is bell-shaped. Approximately 68%, 95% and 99.7% of observations fall within 1, 2 and 3 standard deviations.

Module 6: Central Limit Theorem

The Central Limit Theorem states that the sampling distribution of the sample mean approaches a normal distribution as sample size becomes sufficiently large (commonly $n \geq 30$), regardless of the original population distribution, assuming independent observations and finite variance.

Example

Repeatedly taking samples of size 40 from a skewed population produces sample means that are approximately normally distributed.

Module 7: Confidence Intervals

A confidence interval provides a plausible range for a population parameter.

Module 8: Hypothesis Testing

Use the null hypothesis, alternative hypothesis, significance level and p-value to make statistical decisions.

Interview Questions

1. Explain CLT.
2. Difference between variance and standard deviation.
3. What is a p-value?
4. When should you use a Z-test?
5. Explain confidence intervals.

Summary

CLT is fundamental because it enables statistical inference using sample means.